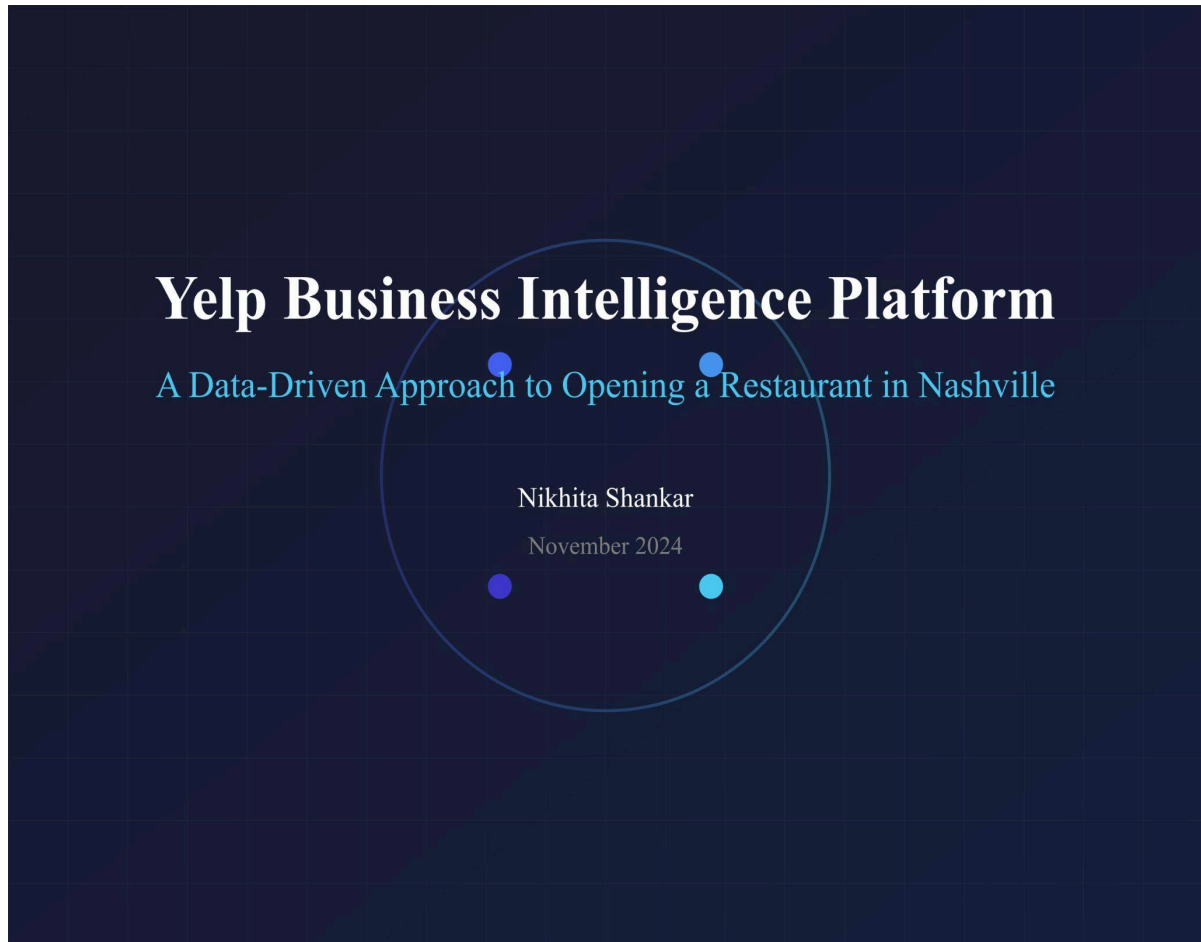




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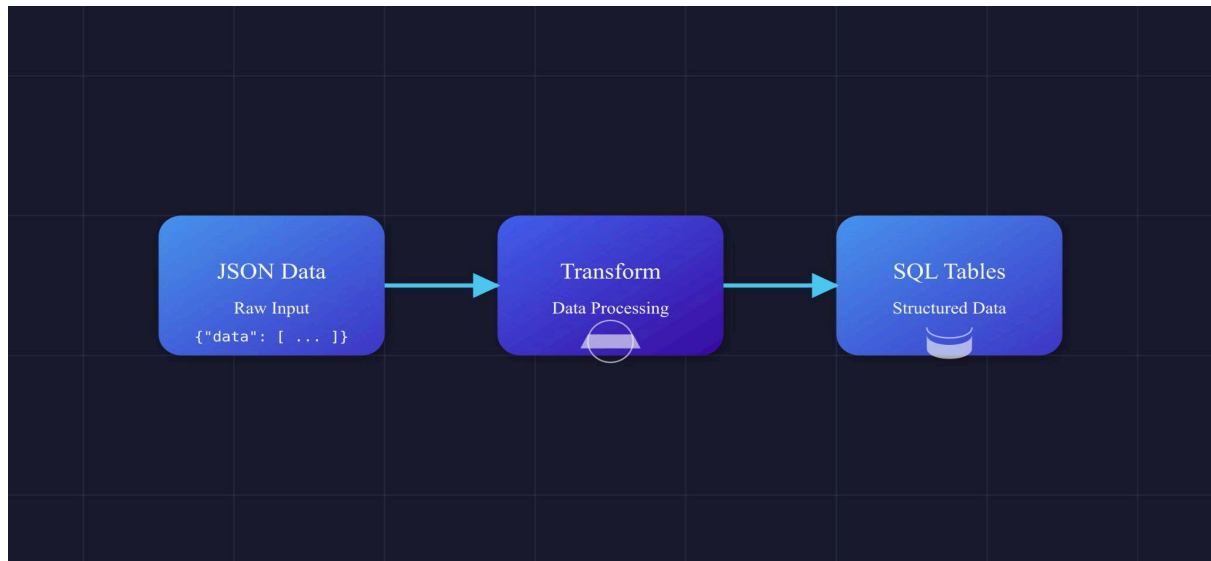
### 1. Introduction

This project focuses on converting Yelp's JSON-format data into structured SQL Server (Azure) data and deriving insights to help set up a new restaurant based on customer data. The primary goal was to explore the dataset, clean and transform the data, and extract valuable business insights using SQL queries.

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### 2. Data Pipeline Overview

- The project utilized Python libraries such as pandas, json, and sqlalchemy to read and process JSON data.
- The data was cleaned, missing values handled, and loaded into structured tables in Azure SQL Server.
- Metadata was specified according to the dataset schema.



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### 3. Technical Stack and SQL Integration

- **Data Exploration:** Python (Pandas, NumPy)
- **Data Conversion:** SQLAlchemy, Azure SQL Server (for creating and managing tables)
- **Visualizations:** Matplotlib, Seaborn

The structured data allowed queries such as identifying top-rated cities, popular cuisines, and the busiest times for restaurants.

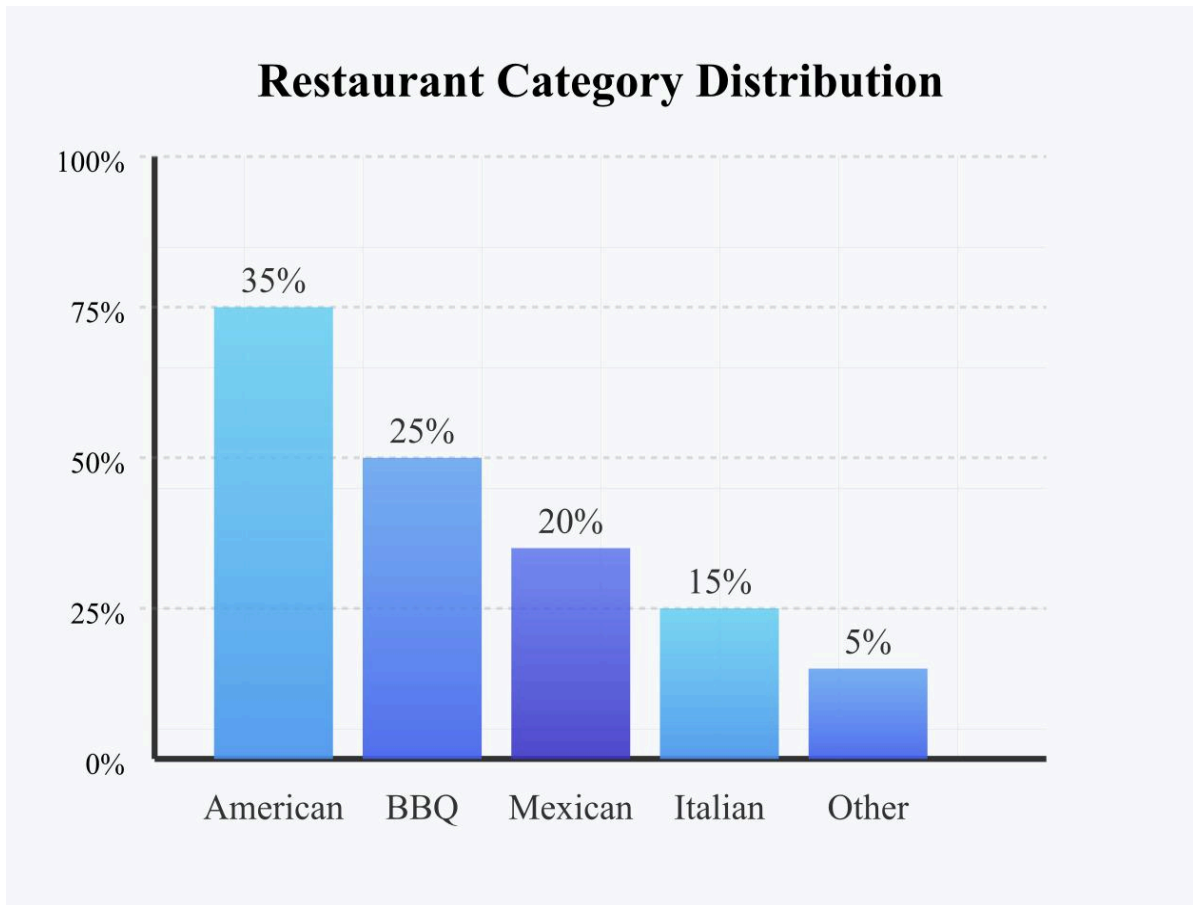


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### 4. Exploratory Insights

- **Restaurant Categories:** Categories such as "American," "Fast Food," and "Seafood" have high representation.
- **Popular Days and Times:** Weekends show peak customer engagement.

- **Customer Sentiment:** The average review rating is 3.75 stars, with most reviews rating 4-5 stars.



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## 5. Business Recommendations

Based on the insights gained, it is recommended to:

1. Capitalize on customer engagement trends by offering promotions on weekends.
2. Focus on maintaining quality, as higher-rated restaurants tend to attract more customers.
3. Introduce menu options that align with popular categories, such as American and BBQ.

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## 6. Why Nashville?

- Suppose you are looking to open a new restaurant or choose a place to dine in based on rich data insights. After analyzing Yelp's extensive dataset, Nashville stands out as an exciting choice due to its diverse and thriving food scene.

- Nashville ranks among the top 10 cities for restaurant density, with over 2500 listed establishments.
- Its popular categories, including American, BBQ, and Fast Food, reflect a well-rounded food culture that caters to a broad audience.
- The data also shows that weekends experience peak customer engagement, making it an excellent opportunity to offer special promotions or unique experiences.

Choosing Nashville as a new restaurant location makes strategic sense for anyone seeking to capitalize on a well-rated, bustling restaurant environment while offering standout service and cuisine.

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## 7. Conclusion

This project demonstrated the value of using Yelp data for strategic decision-making. The JSON-to-SQL conversion process and data exploration enabled valuable insights for planning a successful restaurant launch.

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## 8. Contact Information

- **Author:** Nikhita Shankar
  - **Email:** [nikhita4@illinois.edu](mailto:nikhita4@illinois.edu)
  - **LinkedIn:** [linkedin.com/in/nikhita-shankar-](https://www.linkedin.com/in/nikhita-shankar-)
  - **GitHub:** [github.com/NikhitaShankar97](https://github.com/NikhitaShankar97)
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